

Labor Market Returns to Socio-Emotional Traits and the Impacts of Training Programs: A Meta-Analysis^{*†}

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March 4, 2026

Abstract

This paper synthesizes the growing evidence on the labor market returns to socio-emotional skills and the effectiveness of training programs. While existing studies document elevated returns to socio-emotional skills in the labor market, there is little consensus on which specific aspects of the skills drive these gains. Utilizing one of the established psychological measures, the Big Five Inventory (BFI) personality traits, we define relevant keywords for socio-emotional skills, web-scrape literature, and construct coefficients database on the returns and training impacts. DerSimonian and Laird random-effects estimates suggest that a one standard deviation increase in each of BFI is associated with a 1.4 to 3.1 percentage point increase in wages, with substantial heterogeneity across individual traits, demographic groups, and country income levels; among BFIs, ‘disagreeableness’ exhibits the highest wage returns (3.1 percentage points). Training programs improve BFI by up to 0.46 standard deviations (SD) overall, with larger effects for attitude-related traits such as openness (0.667 SD) and conscientiousness (0.400 SD). Although early childhood constitutes a critical period for cognitive skill formation, our evidence indicates that socio-emotional competencies remain markedly malleable in later developmental stages. Well-designed post-secondary interventions can still generate substantial gains.

JEL Codes: I26, J24, J31

Keywords– Socio-emotional Skills, Labor Market Returns, Skills Training, Big Five Inventory (BFI), Quantitative Meta-Analysis

^{*}The authors are grateful to Wendy Cunningham, Victoria Levin, Harry Patrinos, Yevgeniya Savchenko, Jonathan Stoterau for their review of the paper, and Luis Eduardo San Martin and Philipp Zimmer for their methodological review. The paper benefited from comments and discussions at the World Bank Economics of Education Seminar Series, Southern Economic Association Conference, and Public Sector Economics Conference.

[†]Disclaimer: The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors and do not necessarily represent the views of the International Bank for Reconstruction and Development/World Bank and its affiliated organizations, or those of the Executive Directors of the World Bank or the governments they represent. All errors are our own.

1 Introduction

The value of skills in the labor market has shifted dramatically over the past century. In the United States, for instance, the wage gap between high-skilled and low-skilled workers has continued to widen (Katz and Murphy, 1992). Skill-Biased Technical Change (SBTC) drove this trend in the late 1990s and early 2000s. The earnings of the top percentile and bottom percentile have diverged due to an increasing skill technology bias that cannot be easily automated or off-shored (Autor et al., 2003; Acemoglu and Autor, 2011). However, with further technological advancements, particularly with the emergence of artificial intelligence (AI), more recent studies suggest that returns to cognitive skills in the labor market are beginning to decline (Castex and Kogan Dechter, 2014; Deming, 2017; Ashworth et al., 2021; Hellerstein et al., 2024). This shift places specialized tasks, such as legal analysis and basic medical prescriptions, at risk (Castex and Kogan Dechter, 2014; Brynjolfsson and McAfee, 2014; Beaudry et al., 2016; Edin et al., 2022). Initially observed in advanced economies, this change has also been noted in the labor markets of other countries, albeit to varying degrees, influenced by contextual, institutional, and educational factors (Deming and Silliman, 2024).

The diminishing returns to cognitive skills have been accompanied by rising relative returns to workers’ socio-emotional skills. For example, Deming (2017) shows that a young worker switching to a job requiring 10% higher social skill intensity experienced a 2.1 percentage point wage increase (Deming, 2017). Meanwhile, the wage gain from switching to a more math-intensive job declined by 0.9 percentage points in 1997 compared to 1979. More recent evidence suggests that returns to socio-emotional skills and traits rapidly grew from the late 1990s to the early 2000s and continued growing thereafter (Edin et al., 2022). Articles commonly stress that workers with strong socio-emotional abilities sort into tasks in which they have a comparative advantage, and the increasing value of socio-emotional skills in labor markets elevate their wage returns.

In light of their growing significance, scholars argue that the developmental support of socio-emotional skills is crucial to mitigate inequality in these skills. Studies find that socioeconomic disparities contribute to differences in socio-emotional skills among children (Attanasio et al., 2020), which can perpetuate a vicious cycle of wage gaps in the next generation. However, a silver lining is that socio-emotional skills and traits are malleable until later stages of life, allowing for gradual yet continuous improvements across the life course (Atherton et al., 2021; Graham and Flamini, 2023).

Indeed, there is a substantial body of experimental and quasi-experimental studies highlighting the effectiveness of socio-emotional skills training at different life stages. Gertler et al. (2014) found that parental interventions that included psychosocial stimulation for toddlers in Jamaica resulted in more than a 25 percent increase in earnings 20 years later. Allemand et al. (2023) demonstrated that training in conscientiousness for workers in Senegal’s rail construction sector led to an immediate income increase of 22.7 percent. The evidence regarding the context and timing of interventions for socio-emotional skills development and their outcomes is diverse and expanding (Gertler et al., 2014; Premand et al., 2016; Rammstedt et al., 2024).

Motivated by the growing importance of socio-emotional skills and the exponential growth of the related literature, this study synthesizes recent empirical evidence on (i) the wage returns to workers’ socio-emotional skills and (ii) the effectiveness of training programs aimed at cultivating the traits. A central challenge is how to define ‘socio-emotional skills.’ We adopt the Big Five Inventory (BFI) Personality Traits (Agreeableness, Conscientiousness, Extraversion, Neuroticism, and Openness) as our analytical framework, given its widespread use and its direct relevance for

skills-training interventions. In terms of representation, each BFI trait can be translated into broader socio-emotional skill keywords. For instance, the Harvard EASEL Taxonomy Project recently mapped the Big Five to hundreds of related socio-emotional skill keywords (Jones et al., 2019).

For the analysis of returns, we first synthesize the return coefficients reported as the percentage change in a worker’s real wage from recently published journal articles and working papers. Until March 2025, there were 17,500 resources in Google Scholar with the term ‘socio-emotional skills’ in the title or abstract. Through the systematic and replicable process of Boolean keyword-based web scraping and content review with the assistance of Generative Pre-Trained Transformer (GPT) Artificial Intelligence (AI), we narrowed them down to relevant studies. There were 1,896 coefficients on the returns to cognitive and socio-emotional skills, with a few foundational papers released earlier. Among them, we conducted a final extraction of 613 relevant coefficients from 16 papers through human screening.¹

The analysis shows that a one standard deviation (SD) increase in any BFI trait is associated with a 1.4 to 3.1 percentage point increase in wages, roughly one-third of the reported 10 percent return to cognitive skills, often proxied as one additional year of schooling (Card, 1999). Interestingly, the heterogeneity across the BFIs is moderate. Among the traits, disagreeableness yields the highest return, with a 3.1 percent point increase in wages, followed by Emotional Stability and Openness, each with returns at 2.5 and 2.4 percent points. Conscientiousness and Extraversion contribute more modestly. In subsample analyses, we find substantial heterogeneity across educational attainment, gender, and country income levels, with the highest returns observed in high-income countries. However, most available evidence in developing countries are from Latin America, as of March 2025, highlighting the need for additional evidence from more diverse countries.

Next, we investigate the effectiveness of training programs aimed at improving socio-emotional skills. Our analysis focuses on whether these skills can be developed through structured interventions and, if so, which program designs are most effective. Given the overwhelmingly large volume of training literature, much of which includes socio-emotional skill components yet mixed with cognitive skills or social-protection aspects, we adopted more discretionary and targeted selection schemes to avoid results confounding. In the current version of the manuscript, we conducted a web scraping of earlier meta-analysis papers on socio-emotional and other non-cognitive skills. The initial extraction yielded 3,194 papers. Using a GPT-assisted human screening process, we narrowed this set to 29 relevant meta-analyses, which form the empirical backbone for our program characteristics and impacts analysis.²

Our analysis indicates that socio-emotional skills can be increased by as much as 0.46 standard deviations from trainings. Programs that target attitude-related traits, especially openness (0.67 SD) and conscientiousness (0.40 SD), achieve the largest gains compared to interventions focused on personality or emotional traits such as extraversion (0.38 SD), emotional stability (0.36 SD), and disagreeableness (0.16 SD). However, the differences among conscientiousness, extraversion, and emotional stability are not large. In subsample analysis by region, we found that disagreeableness is less uniformly rewarded in labor markets. Training offered early in life, when skills are most

¹To avoid bias from dependent variable conversion, we restricted our analysis to log real wage estimates. The final selection of 16 papers results from excluding studies that employ nominal wages or other adjacent earnings definitions as the dependent variable, as they cover different countries and years.

²While this approach allows us to extract the most topic relevant evidence, relying on meta-analyses raises several concerns such as literature redundancy and the handling of weights in pooled estimates. In the next version of the paper, we will address these issues by reporting the distribution of estimates under alternative weighting schemes.

malleable, has the greatest impact. However, programs delivered at later stages still generate meaningful improvements, showing that socio-emotional skills remain amenable to development across the life course.

The findings on training also suggest that traditional approaches, direct instruction and inquiry-based pedagogy, are especially effective, whereas technology-reliant methods generate smaller gains. Notably, programs delivered by non-teaching staff outperform those led by regular teachers, implying that non-faculty instructors can add particular value in building socio-emotional skills. Broad, general-audience programs yield the largest impact while targeted programs show smaller impacts, possibly because they often serve more challenging populations.

This paper is organized as follows. Section 2 explores the definition and taxonomy of skills in academic disciplines and policy frameworks, focusing on socio-emotional skills and traits. Section 3 introduces existing measures for assessing socio-emotional skills, labor market returns, and program interventions. In this section, we also describe the measurements and framework adopted in our study. Section 4 explains our human-machine collaboration process for constructing the literature database, and section 5 discusses our estimation strategy for conducting the meta-analysis. Section 6 presents our analytical results on the returns to socio-emotional skills and the training impacts. Finally, section 7 summarizes the key takeaways from our study and suggests additional directions of investigation.

2 Skills Taxonomy

As highlighted in Heckman (2006), the structure of the brain and the process of skill development are shaped by the interplay between genetics and personal experiences. Skills are often understood as ‘a special ability to do something, developed through training, experience, or practice’ that enables a person to perform specific tasks or activities effectively. On the other hand, traits are inherent characteristics or qualities that define an individual’s personality (John et al., 2008). Both skills and traits are important determinants of a person’s socio-emotional characteristics. Collectively, cognitive and socio-emotional abilities are interconnected and contribute to educational and labor market success, determined by the market’s supply and demand for these abilities.

Due to the complex developmental interplay and subsequently varying definitions, the skills taxonomy also differs across disciplines and policy entities. Education scholars often categorize skills into cognitive, linguistic, and socio-emotional competencies (Heckman, 2006; Hilton and Pellegrino, 2012). These approaches often benchmark Bloom’s classification of cognitive, affective, and psychomotor learning goals (Bloom, 1974). In contrast, occupation- and task-based approaches classify skills by functionality. For instance, the content model of the United States Occupational Information Network (ONET) divides basic skills into static content and procedural skills. Using open systems theory as a benchmark (Katz and Kahn, 1978), the ONET further classifies more advanced cross-functional skills into problem-solving, resource management, social, systems, and technical skills (Peterson et al., 2001). Lastly, policy entities advocate for more structured and staged skills frameworks. For instance, the World Bank skills classification regards skills as key determinants of human development, productivity, and ultimately economic growth (Cunningham and Villaseñor, 2016). Foundational cognitive skills include basic literacy, both language and digital literacy, and numeracy, which are essential for acquiring more advanced skills. Additionally, socio-emotional skills, crucial for managing and regulating relationships, emotions, and attitudes, are also regarded as foundational skills, which serve as an important basis for the learning of higher-level skills. At a

more sophisticated stage, higher-order cognitive skills encompass problem-solving, communication, information analysis, and critical thinking. The final category includes occupation-specific skills, comprising both technical and cognitive abilities tailored to specific job roles, with specialized skills, such as entrepreneurship, also playing a significant role.

With recent evidence of the declining returns to cognitive skills in labor markets (Castex and Kogan Dechter, 2014; Brynjolfsson and McAfee, 2014; Beaudry et al., 2016; Edin et al., 2022), scholarly investigations and policy efforts are shifting toward the non-cognitive aspects of skills. The definition of socio-emotional skills is evolving in articles. Until now, there exists no consensus on how to define and measure them (Whitehurst, 2016). Socio-emotional skills are indicated by various terms, such as non-cognitive skills, personality traits, 21st-century skills, soft skills, and more recently, higher-order non-cognitive skills (Duckworth and Yeager, 2015; Kankaraš, 2017; Cabus et al., 2021; Deming and Silliman, 2024), which is often regarded as a term that encompasses complex traits with overlapping boundaries. Still, the concept is gradually evolving within the discipline of economics (Deming and Silliman, 2024), and recent studies have started to focus on a few key indicators to serve as representative measures.

In the policy domain, several structured frameworks exist for classifying non-cognitive skills, each reflecting distinct measurement priorities, theoretical orientations, and policy applications. The OECD Social and Emotional Skills Framework focuses on developing lifelong competencies necessary for academic, career, and personal success, aligning closely with human capital and well-being indicators (Jones et al., 2019). Meanwhile, the Collaborative for Academic, Social, and Emotional Learning (CASEL) framework structures non-cognitive skills into five core competencies, including self-awareness, self-management, social awareness, relationship skills, and responsible decision-making, primarily for application in education systems and youth development programs.

These frameworks contrast with trait-based models, which measure inherent personality characteristics rather than trainable competencies. However, traits can be mapped onto trainable competencies in the above skill-based frameworks, such as conscientiousness aligning with goal-setting and self-discipline or agreeableness relating to conflict resolution and teamwork. The literature evaluating labor market returns to non-cognitive skills has extensively used the traits model, largely due to its longer history in psychological research and the availability of standardized measures across countries. Moreover, the traits do not suffer from the bias and interests potentially present in more structured frameworks by policy entities. We conclude that the trait models, represented by BFI, offer the most comprehensive academic evidence base for analyzing economic returns.

3 Measuring Socio-Emotional Skills, Returns, and Training Impacts

3.1 Available Measures to Assess Socio-Emotional Skills

Connected to human capital theory, skills have long been measured by educational attainment (Card, 1999, 2001; Psacharopoulos and Patrinos, 2018). The predominant focus in economic literature has been on using years of schooling to analyze skill levels and labor market returns for workers. However, criticism has arisen regarding the reliance on years of schooling as a measure, largely due to variations in educational quality and environment across countries and regions. Additionally, many skills are acquired outside formal educational systems (Hanushek and Woessmann, 2008; Cunningham and Villaseñor, 2016). Furthermore, using educational attainment as a measure

is somewhat crude, as it fails to clearly distinguish between cognitive and non-cognitive aspects of skills. New metrics such as Learning-Adjusted Years of Schooling (LAYS) have been developed to better reflect actual learning outcomes rather than mere time spent in school (Filmer et al., 2020). These measures have been incorporated into the World Bank’s Human Capital Index, helping shift the global focus from schooling quantity to education quality (skills gained) as a key driver of economic development (Hanushek, 2013). However, the limitations, particularly in measuring socio-emotional skills, remain persistent.

Statistical agencies and human resources firms publish skill categories and competency scales, linked to occupations and tasks. A growing body of literature examines skills as vectors that constitute tasks and occupations. Research on labor migration, tasks, and occupations frequently utilizes these measures (Yamaguchi, 2012; Autor and Handel, 2013; Robinson, 2018; Acemoglu and Restrepo, 2020). However, the task-based classifications are not well harmonized across diverse public and private entities in different countries. The lack of standardization introduces subjectivity when constructing task-skill vectors. From the perspective of socio-emotional skills, the measures tend to be more occupation- and task-centric, often failing to capture unique features of non-cognitive nature, such as their latent development (Lavado et al., 2021).

To standardize the measurement of skills, scholars have turned to widely recognized assessment tools. For example, direct task- and test-based assessments like the Programme for International Student Assessment (PISA) and the Survey of Adult Skills (PIAAC) provide generalizable metrics for basic skills such as literacy, numeracy, and digital skills. While these assessments excel in measuring and comparing cognitive skills using task-based robust psychometric tools, they fall short in measuring the socio-emotional factors that influence non-cognitive skills and worker productivity, which are often collected through self-reported questionnaires.

Structural estimation efforts represent another approach to measuring skills. The structural approach views skills as latent variables rather than directly observable ones (Keane and Wolpin, 1997; Sarzosa and Urzúa, 2016), acknowledging that observed skills data may be particularly prone to measurement error. Researchers employing this approach use structural estimation methods to interpret survey results and test scores as outcomes of the latent skills they estimate through their models. Until recently, most studies using the structural approach have focused primarily on learning skills, such as reading comprehension (Carneiro et al., 2003). Socio-emotional traits, such as extraversion and openness, had not been incorporated in this approach, despite some structural evidence of their interplay with learning skills (Acosta et al., 2020). A new approach suggested by Deming and Silliman (2024) proposes that non-cognitive or “higher-order” skills can be integrated into the human capital model as a component of task choice. This latent variable conceptualization is gradually incorporating non-cognitive dimensions.

Lastly, personality trait measures constitute another widely used approach to measuring skills, particularly in health and psychological domains. Sets of personality measures have gradually developed from the early 1900s through the 1980s, with the Big Five (BFI) traits serving as foundational concepts for many subsequent evolutions. The five traits are agreeableness, conscientiousness, extraversion, neuroticism, and openness, and related keywords can be mapped to broader taxonomies.

In this paper, we employ the BFI personality traits due to their broader keywords adaptability and neutrality across diverse policy frameworks and occupational classifications. Table 1 describes each personality trait and the associated keywords used in our analysis. As elaborated in the table, these socio-emotional skills often pertain to the management of attitudes, relationships, and emotions, as well as to higher-order cognitive skills such as critical thinking and problem solving.

3.2 Big Five Personality Traits as Socio-Emotional Skill Measures

Table 1: Mapping of Non-Cognitive Skills to Relevant Domains and Key Words

Big Five Traits	Related Domain	Key Words
Neuroticism	Management of attitudes	Emotional regulation, Stress management, Mindfulness
Conscientiousness	Management of attitudes	Self-control, Self-discipline, Goal-setting, Impulsivity, Diligence
Agreeableness	Management of emotions	Assertiveness, Conflict resolution, Empathy, Bystander Intervention
Openness	Higher-order cognitive skills	Creative thinking, Flexibility, Adaptability, Resilience, Problem solving
Extraversion	Management of relationships	Social skills, Communication skills, Collaboration, Teamwork

Many studies employ BFI (John et al., 1991) as proxies for non-cognitive skills. These indicators separately measure basic socio-emotional skills (e.g., agreeableness) and higher-order skills (e.g., conflict resolution) without concerns of confounding, and they are assessed using standardized questionnaires, thereby reducing subjectivity.

Despite the benefits, limitations exist in measuring non-cognitive skills with the BFI indicators. First, the Big Five Traits are typically measured using self-reported answers, which introduces subjectivity and the potential for social desirability bias (see the Big Five Inventory Traits Questionnaire in Figure A6). Respondents may unintentionally exaggerate desirable traits like conscientiousness or downplay less favorable traits such as neuroticism (Schmitt et al., 2007). Additionally, situational factors, such as mood at the time of assessment or cultural influences, can affect responses (Podsakoff et al., 2003; Schmitt et al., 2007). The estimates in this paper should be interpreted cautiously with these two limitations (Almlund et al., 2011).

Many studies supplement or validate the Big Five traits with additional characteristics. For instance, the HEXACO Personal Inventory expands on the Big Five by including the H-Factor (Lee and Ashton, 2004; Cabus et al., 2021). Other variables include decision-making (Balcar, 2016), communication (Balcar, 2016; Acosta et al., 2020), cooperation (Balcar, 2016), emotional stability (Cunningham and Villaseñor, 2016; Acosta et al., 2020), grit (Díaz et al., 2012; Danner et al., 2019; Acosta et al., 2020), hostile attribution bias (Acosta et al., 2020), locus of control (Heineck and Anger, 2010; Bühler et al., 2020a), reading proficiency (Acosta et al., 2020), reciprocity (Heineck and Anger, 2010), risk preference (Bühler et al., 2020a), patience (Bühler et al., 2020a), perseverance (Cunningham and Villaseñor, 2016), self-efficacy (Lavado et al., 2021), self-esteem (Lavado et al., 2021), and trust (Acosta et al., 2020).

Accordingly, this meta-analysis compiles return and training impact estimates primarily based on the BFI measures but also connecting them to broader adjacent keywords. By standardizing the unit of analysis to represent a one SD increase in skills and collating hundreds of coefficients for each of the five personality traits, we summarize recent study findings on socio-emotional skill returns and training impacts.

3.3 Measuring Labor Market Returns

In this paper, we primarily employ log real monthly wages to investigate the returns to non-cognitive skills, imposing sample restrictions to enhance standardization and minimize variation arising from measurement differences. This choice reflects the fact that converting wages reported in different units, such as nominal wages in local currencies, can introduce bias when harmonizing values across time and countries. Moreover, the existing body of evidence on log real wages is sufficiently large to support robust inference in a meta-analysis without introducing additional measurement error from purchasing power parity adjustments. Therefore, this study builds its evidence base on the narrowest outcome variable, which minimizes bias and provides a cornerstone for broader analysis.

Wage outcomes may not fully capture the value of non-cognitive skills, which can also affect other labor market outcomes. Considering other indicators, employment cannot be overlooked (Adoho et al., 2014; Ibarraran et al., 2014; Cunningham and Villaseñor, 2016; Lavado et al., 2021; Schultheiss and Backes-Gellner, 2023). From the simple measure of comparing the employment rate of workers with varying non-cognitive skills, the job landing among unemployed trainees and the likelihood of starting business among entrepreneurs all fall into this category. Other indicators employed by recent studies are, career advancement (Adhvaryu et al., 2023), contract type (Cunningham and Villaseñor, 2016), educational investment (Acosta et al., 2020; Lavado et al., 2021), job satisfaction (Danner et al., 2019), occupational choice (Lavado et al., 2021), saving and investment behavior (Adoho et al., 2014), skill requirement changes for jobs (Allen et al., 2020), tertiary educational attainment (Acosta et al., 2020), the type of occupation (Cunningham and Villaseñor, 2016; Acosta et al., 2020). Studies in developing economies also investigated job formalization and other workplace quality indicators as the labor market is not as much institutionalized as in advanced economies (Ibarraran et al., 2014; Cunningham and Villaseñor, 2016; Acosta et al., 2020). Future research could systematically evaluate these additional labor market effects.

3.4 Classifying Training Interventions

Socio-emotional skills, like cognitive skills, were often considered less malleable as individuals age. However, recent studies indicate that these traits can undergo small but consistent changes throughout the life cycle, depending on a person’s living environment and skills development nudges (Atherton et al., 2021). Indeed, existing evidence shows that socio-emotional skills training can be highly effective for adult trainees. The effectiveness of programs vary by implementation modalities, target populations and regions, as well as the trainees’ life stages.

Childhood Intervention During early childhood, modalities such as play-based learning, teacher training, and parental engagement are particularly effective. Programs like Promoting Alternative Thinking Strategies (PATHS) have been found to effectively reduce problem behaviors and improve social and academic behaviors by utilizing interactive teaching strategies to foster skills, such as emotional regulation and cooperation through role-play and group activities (Domitrovich et al., 2007; Group, 1999; Bierman et al., 2010). Additionally, parental involvement in reinforcing these skills at home has been shown to accelerate child development, though it is often more challenging in low-resource environments (Luthar and Eisenberg, 2017). Teacher training, combined with structured curricula, ensures that socio-emotional skills are consistently practiced in educational settings, leading to long-term benefits that extend into adolescence and adulthood. In addition

to structured interventions, the use of professional development programs like Social Emotional Engagement-Knowledge & Skills-Early Childhood provides educators with the skills necessary to foster children’s socio-emotional development. These programs have shown positive outcomes in enhancing social-communication skills in children, particularly those with challenges in this area (Siller et al., 2024).

Intervention for Adolescents and Adults. Adolescence is a critical period for shaping socio-emotional skills that are strongly linked to long-term socio-economic success. As children transition into adolescence, the focus of interventions shifts towards cognitive and social skills training. For instance, cognitive behavioral therapy (CBT) and social skills training have been used to improve self-control, communication, and resilience, particularly in at-risk youth. These interventions are most effective when paired with secure, supportive environments that mimic successful family structures, allowing adolescents to develop these skills within safe boundaries (Kautz et al., 2014).

However, several challenges persist in designing and implementing these programs. One key issue is the variability in individual responses to interventions. Factors such as age, cultural background, and personal circumstances significantly influence the effectiveness of socio-emotional skill training. Interventions must account for these differences to ensure their applicability across diverse populations, particularly in settings where cultural norms may affect how traits like conscientiousness or agreeableness are developed and expressed (Laajaj et al., 2019).

Despite these challenges, there are significant opportunities for enhancing socio-emotional skills through well-designed interventions. Research has consistently demonstrated strong links between traits such as self-control, cooperation, and emotional regulation and positive life outcomes, including academic success, career achievement, and social engagement (John et al., 2020; Stahlmann et al., 2024). Programs that target these traits during adolescence and early adulthood can produce lasting benefits. Moreover, key life transitions, such as moving from school to work, entering leadership roles, or retirement—offer unique opportunities for targeted interventions (Kornadt et al., 2019). For example, leadership training during the transition to higher education or the workforce can enhance traits like openness and extraversion, leading to improved performance and social outcomes.

Given the malleability of socio-emotional skills and the opportunities to enhance them across various stages of life, targeted training programs offer a promising avenue for development. This study focuses on evaluating the effectiveness of such interventions, particularly those designed to improve BFI for adolescence in secondary education and early adulthood in post-secondary education and above. By synthesizing findings from existing meta-analyses, we aim to identify patterns and examine heterogeneity across key moderators, such as school grades, intervention setting, instructor characteristics, and population targeting. This approach not only leverages a vast body of evidence efficiently but also provides a nuanced understanding of how different modalities and features of training programs influence the program outcomes. Focusing on school- and workplace-based interventions, we have compiled evidence to form a comprehensive literature database of existing meta-analyses. The next section outlines the methodologies and processes used to construct the meta-analytic literature database.

4 Database Construction

This study employs a researcher-machine collaborative protocol to construct a comprehensive literature database on the returns to socio-emotional skills and the impacts of training programs. Using Boolean keyword-based web scraping of Google Scholar, we initiate a systematic literature search that identifies 161 skill return papers and 3,194 skill training papers published between 2019 and 2024, supplemented with a small number of earlier benchmark references. We then train a Generative Pre-trained Transformer (GPT) model for structured information extraction and apply rigorous human screening. This workflow enables us to efficiently screen and identify relevant papers.

For wage return estimates, the study reduces an initial set of 190 papers containing 1,896 coefficients to 16 papers with 613 effect sizes, including 568 estimates for socio-emotional skills and 45 for cognitive skills. While this narrow evidence base may not fully capture the broader literature on the wage effects of socio-emotional skills, particularly beyond BFI and log real monthly wage measures, we prioritize close harmonization and comparability of outcomes. This approach enhances internal consistency and precision, providing a robust foundation for estimation and results interpretation.

To estimate the impact of training programs, we conducted evidence synthesis on meta-analyses. Due to overwhelmingly large literature with mixed evidences on socio-emotional skills training, we focused on extracting meta-analysis papers rather than the individual studies. This approach achieved efficiency by allowing us to leverage the rigor of existing meta-analysis in screening training literature. By synthesizing meta-analyses, we identified patterns and addressed variability in factors such as student grade levels, modalities, and population targeting. The initial search yielded 3,194 training papers. However, after applying the narrower and rigorous screening process based on set of inclusion and exclusion criteria in the Appendix A4, we finalized a list of 29 meta-analyses with 364 coefficients, focusing on interventions designed to enhance socio-emotional skills.

To estimate the impacts of training programs, we conduct an evidence synthesis of existing meta-analyses. Given the exceptionally large and heterogeneous literature with mixed program components and evidences on socio-emotional skills training, our study web-scrapes meta-analysis papers rather than individual studies. This strategy improves efficiency by leveraging the rigorous screening, aggregation, and quality controls embedded in prior meta-analyses. Synthesizing meta-analyses allows us to identify systematic patterns while addressing variation across key dimensions, including student grade levels, delivery modalities, and target populations. Our initial search identifies 3,194 training-related papers. Applying a narrower and more rigorous screening process—based on predefined inclusion and exclusion criteria detailed in Appendix Figure A4, we retain 29 meta-analyses, comprising 364 effect-size estimates, all of which evaluate interventions explicitly designed to enhance socio-emotional skills.³

We conducted the literature web-scraping on the returns to socio-emotional skills from March 26th, 2024, until May 15th, 2024, and the training impacts from September 1st, 2024, until October 31st, 2024. The detailed web-scraping, machine training, and database construction process such as inclusion and exclusion criteria are elaborated in the Appendix A.

³While meta-analysis paper scraping improves efficiency, it raises several concerns, including potential evidence double counting and uneven weighting of estimates. In future versions of the paper, we will present distributions of estimates using alternative weighting schemes.

5 Methodology

With the narrowly synthesized set of evidence, we conducted the Panel Random Effects (RE) estimation known as DerSimonian and Laird methods (DerSimonian and Laird, 2015). This approach was chosen over Fixed Effects (FE) meta-analysis methods because it allows for the effect sizes to vary across studies rather than assuming they are identical as in the FE methods. The flexibility of RE is particularly appropriate in this setting, as the studies included in the analysis differ in their sample sizes, target populations, model specifications, and the measurement scales of socio-emotional skills, although our data collection protocol narrowly synthesizes the evidence as possible. While the dependent variable, log real monthly wage, provides a percentage interpretation for all our selection of literature, variations in scaling and methodological choices across the literature necessitate an estimation method that can account for heterogeneity.

The RE model addresses potential sources of variance both within and across studies by assigning smaller weights to studies with significantly different effect sizes. As shown by the formula below, the weighting mechanism minimizes estimation bias arising from varying research structures while better representing more common components across studies. By reducing the influence of outliers, the RE method enhances the robustness of the analysis and provides a more generalized estimate of the returns to socio-emotional skills. While the DerSimonian and Laird estimator can introduce publication bias when the number of studies is limited (Guolo and Varin, 2017), the hundreds of estimates included in our analysis provide sufficient power to mitigate this concern.

$$w_i = \frac{1}{\sigma_i^2 + \tau^2} \quad (1)$$

Although the RE model was the primary estimation strategy due to its suitability for synthesizing heterogeneous evidence, we conducted robustness checks using FE meta-analysis. The FE approach served as a benchmark to confirm the consistency of results under the assumption of uniform effect sizes. Additionally, we identified the preferred model from each paper and performed another robustness check using the RE model on this narrower pool of effect sizes. However, we interpret these estimates with caution, recognizing that a limited number of effect sizes may pose challenges for the DerSimonian and Laird estimator.

To compute the pooled effect size ($\hat{\mu}_j$) for a given socio-emotional skill j , we apply the following weighted average formula:

$$\hat{\mu}_j = \frac{\sum w_{ij} ES_{ij}}{\sum w_{ij}} \quad (2)$$

where w_{ij} is the inverse-variance weight for study i related to skill j , as defined in (1), and ES_{ij} represents the effect size for study i corresponding to skill j .

6 Results

6.1 Labor Market Returns to Socio-Emotional Skills

Descriptive Statistics

Table 2 presents the descriptive statistics of the 613 coefficients from our labor returns analysis, divided into studies examining high-income countries (HICs) and those focusing on developing

countries. Studies on high-income countries (HICs) account for slightly larger portion, with 359 out of 613 coefficients, while studies on developing countries consist of 254 estimates.

For both groups, about 93% of the estimates we collated are on BFIs with the remaining 7% on cognitive skills. Studies on developing countries have a slightly higher share (93.3%) compared to those on HICs (92.2%). The inclusion of education as a control variable varies substantially. In studies on developing countries, 73.2% of estimates include education level as controls, compared to only 28.4% in studies on HICs. This reveals a greater emphasis in developing country contexts on understanding how education interacts with skill effects, while studies on HICs may focus more on standalone skill effects. It is worth noting that most of the studies on developing countries in our sample are from Latin America; thus, caution is warranted in generalizing these results to the entire developing world.

Gender focus shows another stark differences. In studies on HICs, 37.3% and 37.6% of estimates pertain to female and male samples, respectively, with 25.1% analyzing mixed samples. In contrast, 96.1% of estimates in studies on developing countries use mixed samples, with only 2.0% focusing on female or male-specific effects.

Finally, methodological differences are evident between the two groups. Studies on developing countries rely more heavily on instrumental variable (IV) methods (73.6%) compared to studies on HICs (69.9%). Meanwhile, ordinary least squares (OLS) methods are used slightly more frequently in studies on HICs (30.1%) than in those on developing countries (26.4%).

Table 2: Descriptive Statistics, Labor Returns Analysis by Income Group

	All	High income countries	Developing countries
Number of estimates	613	359	254
Estimate type (%)			
Big Five	92.7	92.2	93.3
Cognitive Skills	7.3	7.8	6.7
Education control (%)			
Yes	47.0	28.4	73.2
No	53.0	71.6	26.8
Gender (%)			
Female	22.7	37.3	2.0
Male	22.8	37.6	2.0
Mixed	54.5	25.1	96.1
Methodology (%)			
OLS	28.5	30.1	26.4
IV	71.5	69.9	73.6

Note: This table reports descriptive statistics for the 613 wage-return estimates used in the meta-analysis of labor market returns to socio-emotional skills. The unit of observation is the individual coefficient. The groups of high-income countries (HICs) and developing countries are based on the World Bank income classification at the time of the study. Percentages represent the share of estimates within each characteristic category (estimate type, education control, gender, and methodology).

The Returns to Socio-Emotional Skills

Figure 1 shows the estimation results for the returns to socio-emotional skills with all 613 effects in our dataset. The results indicate that a one SD increase in socio-emotional skills is associated with an approximately 1.4 to 3.1 percentage points increase in log monthly real wages. While this effect is smaller compared to the returns to one year of schooling, the narrow confidence interval suggests that socio-emotional skills do yield statistically significant wage returns to workers. The findings are consistent with existing literature that highlights the generally larger wage returns to schooling or cognitive skills as well as small but evident returns to socio-emotional skills, particularly in settings where educational attainment and technical expertise are prioritized (Heckman et al., 2006).⁴

Among the BFIs, disagreeableness yields the highest returns, with a 1 SD increase associated with a 3.1% increase in wages, followed by emotional stability (2.5%) and openness (2.4%). These findings suggest that disagreeableness may play a significant role in wage determination, particularly in labor markets where negotiation, leadership, or individual competitiveness is valued. The relatively high returns to openness highlight the economic value of creativity, adaptability, and

⁴Recent studies also suggest the complementarity between cognitive and socio-emotional skills, particularly among high-skilled workers, employing skills or task data on cognitive and socio-emotional traits to investigate such complementarity.

intellectual curiosity in modern, dynamic work environments. Similarly, emotional stability, which reflects an individual’s ability to manage stress and remain calm under pressure, is highly valued across occupations that require resilience and effective interpersonal interactions. In contrast, conscientiousness (1.4%) and extraversion (1.1%) have lower but still meaningful returns, indicating the importance of diligence, organization, and sociability in wage outcomes.

Figure 1: The Wage Returns to Big Five Traits vs Cognitive Skills, RE, All Sample

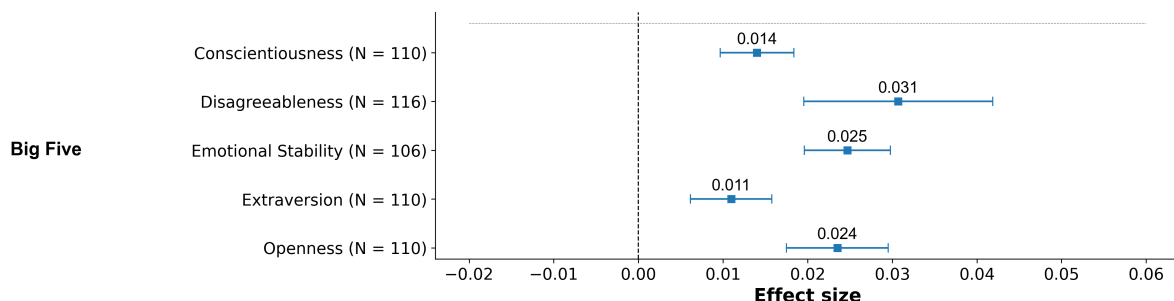
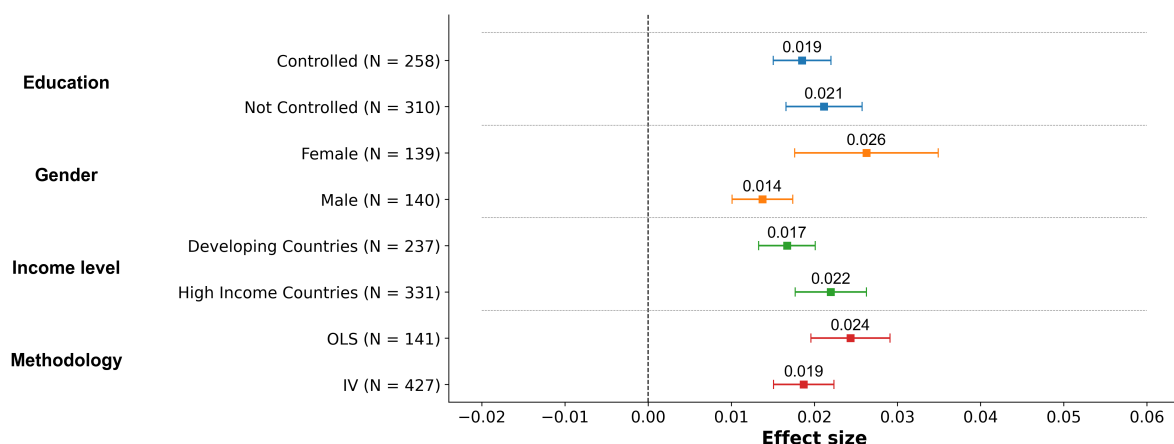


Figure 2: Heterogeneity Effects, RE, All Sample



Note: Figures 1 and 2 report pooled estimates of the labor market returns to socio-emotional skills, measured using the Big Five Personality Traits. Each point estimate represents the average percentage change in real wages associated with a one standard deviation (SD) increase in the corresponding skill. The estimates are based on a DerSimonian–Laird random-effects meta-analysis. Confidence intervals reflect 95 percent uncertainty bands. The total number of estimates included in each pooled effect is reported in parentheses. All estimates are drawn from studies that use log real wages as the dependent variable. In Figure 1, the sum of estimates across individual Big Five traits equals 552. There are 16 estimates that come from one study reporting interaction effects between Big Five traits and tenure. These estimates are included in the overall count (568, excluding 45 cognitive skills coefficients out of 613) but are not assigned to individual traits.

When comparing estimates by country income level, high-income countries demonstrate higher returns, with a 2.2% increase in wages per 1 SD increase in socio-emotional skills, as shown in figure 2. Developing countries show lower returns, at 1.7%. The results suggest that soft skills are slightly more valued in the labor markets of advanced economies than in developing economies.

This difference may also reflect the influence of contextual factors, such as economic structures and cultural norms.

Gender is an important determinant of the returns to socio-emotional skills. The estimated average effect size is higher for women (2.6%) than men (1.4%), and the specific Big Five traits that contribute most to wage increases vary by gender. For women, disagreeableness is associated with the highest returns, with a 7% wage increase per 1 SD increase. This finding aligns with the hypothesis that assertiveness, also interpreted as disagreeableness, may help women navigate traditionally male-dominated labor markets and overcome gender biases in negotiations and leadership opportunities. Conversely, for men, the highest returns are observed for Openness, with a 2% increase per 1 SD increase. This could reflect the premium placed on creativity and intellectual curiosity in male-dominated industries where innovation and exploration are rewarded.

Models with and without education as a control variable yield almost similar results, with returns to a 1 SD increase in BFI Traits estimated at 2.1% when education is not controlled and 1.9% when it is included. The slightly higher returns without controlling for education likely reflect the complementarity between cognitive and socio-emotional personality traits, as schooling can indicate both cognitive and socio-emotional personality traits. For example, it enhances traits like persistence, pro-social behaviors, and emotional regulation. When education is excluded, the estimated returns to socio-emotional personality traits may partially capture the impact of cognitive skills embedded in educational attainment, slightly overstating the effect. However, the modest difference in effect sizes demonstrates that personality traits independently drive wage outcomes, reinforcing their importance in the labor market beyond what is explained by schooling.

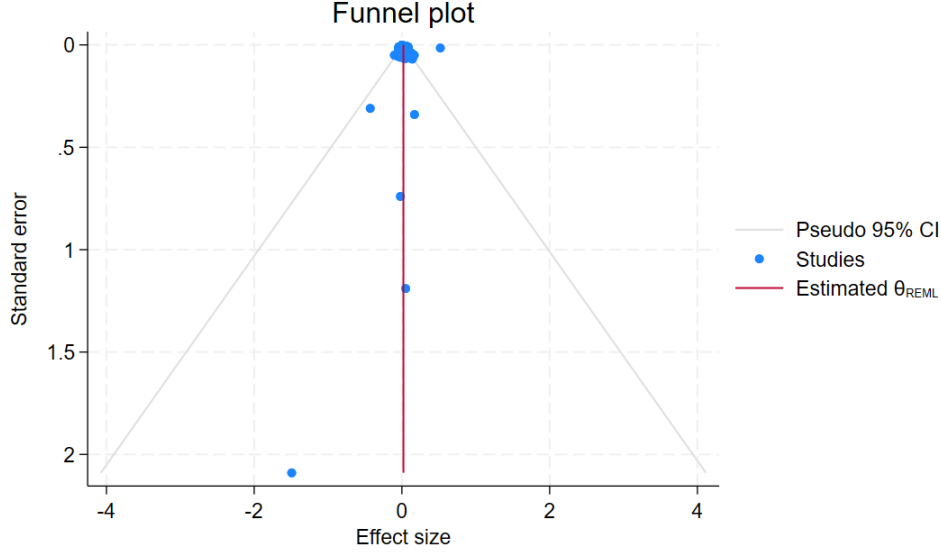
Our estimates are robust across different sample definitions. We pooled effects sizes using only the preferred models identified in each paper. These are models expressed explicitly by authors as their main model, or models that with the most comprehensive set of control variables. The estimates restricted only to the preferred models show less precise estimates as effect sizes were significantly reduced. Figure A7 shows that the average returns to BFI Traits remains in the neighborhood of 2%, while for cognitive skills, the returns were lower at 6%.

We also pooled effect sizes using the fixed effects model as presented in Figure A8. The returns are slightly lower but with greater level of precision. Finally, we selected only the models that included all BFI Traits to ensure the capture of partial effects and to avoid potential bias that may arise from the omission of any trait. This approach allows us to estimate the overall returns to socio-emotional personality traits as the sum of the partial effects of all Big Five traits. The results presented in Figure A9 are consistent with our main findings, reinforcing the robustness of our estimates.

Assessing Publication Bias

To assess publication bias, we use the funnel plot, which is a widely used tool in meta-analysis to visually assess if studies with significant or favorable results are more likely to be published or included in an analysis than those with non-significant results. The funnel plot displays the relationship between the effect sizes of studies and their corresponding standard errors. In the absence of publication bias, the plot should resemble a symmetrical inverted funnel, with the studies with smaller standard errors clustering around the overall effect size at the top and the studies with larger standard errors spreading symmetrically at the bottom. Overall plot asymmetry or the absences of studies with larger standard errors with non-significant results may indicate potential bias.

Figure 3: Returns Estimation: Publication Bias



Note: This figure presents a funnel plot assessing potential publication bias in the estimated labor market returns to socio-emotional skills. Each point represents a study-level effect estimate plotted against its standard error.

For our sample, the funnel plot in Figure 3 shows a mostly symmetrical distribution of studies around the estimated effect size, which suggests that publication bias is minimal. Most studies are clustered near the top of the funnel, reflecting precise estimates from larger sample sizes. However, there is a slight scarcity of studies with non-significant or negative effect sizes in the lower part of the funnel, which could indicate a mild degree of publication bias or simply reflect the challenges in detecting and publishing small, non-significant effects. Additionally, a few studies are outliers, with effect sizes that deviate significantly from the cluster due to heterogeneity in study designs or sample characteristics rather than bias.

6.2 Training Interventions for Socio-Emotional Skills Development

Our estimates in Figure 4 show that training interventions aimed at developing socio-emotional skills have an average effect size of 0.46 SD. The revealed effect suggests that training interventions can be effective in developing socio-emotional skills to realize their returns. Summary statistics describing the characteristics of the corresponding estimates are provided in Table 3

However, there is significant heterogeneity in effectiveness across the BFI traits (Figure 5). Interventions are most effective for openness ($\theta = 0.65$) and conscientiousness ($\theta = 0.40$), with moderate effects observed for emotional stability ($\theta = 0.37$) and extraversion ($\theta = 0.38$). In contrast, the effect size for disagreeableness ($\theta = 0.16$) is notably lower, suggesting that traits such as empathy and standing up against others may be more resistant to change through short-term training. In subsample analysis by region, we found that disagreeableness is less uniformly rewarded in labor markets, with some countries less preferring the quality which may have affected the generally small training impacts.

Overall, these findings highlight that while socio-emotional skill interventions can yield signifi-

cant improvements, traits like openness and conscientiousness appear to be particularly responsive to training efforts. It may be easier to train individuals to manage their attitudes than to manage relationship or to control emotions. Developing disagreeableness may not be ideal for all country contexts, and if desirable, require longer-term or more immersive interventions.

Effects by Educational Level

The effectiveness of socio-emotional interventions varies significantly by educational level. The greatest benefits are realized for adolescents enrolled in the secondary education ($\theta = 0.58$), closely followed by children in primary education or less ($\theta = 0.55$). The findings highlight the significance of early interventions when children's social skills are malleable. For these group, training could be a successful mechanism to develop foundational socio-emotional skills like emotional control and self-discipline (Duckworth et al., 2014).

At the post-secondary level and above, the impact is slightly lower ($\theta = 0.41$). While socio-emotional skills remain malleable throughout life stages, several factors may contribute to reduced effectiveness at this stage. Educational and training priorities at the post-secondary level and above tend to focus more on cognitive skills such as critical thinking, technical expertise, and subject knowledge, and immediately applicable skills for workplace tasks. Academic and career success is largely determined by exams and mastery of content, which prioritize cognitive development over socio-emotional traits. socio-emotional skills like perseverance, adaptability, or emotional regulation are important but often operate in the background, supporting cognitive skill acquisition rather than being directly evaluated.

The findings by educational levels suggests a need for post-secondary institutions to rethink how they approach the development of socio-emotional skills. Currently, academic and workplace performance metrics in early adulthood focus on cognitive competencies, such as exams, projects, and task accuracy and speeds, which reward knowledge retention and critical thinking but often fail to address the importance of socio-emotional skills. To address this, colleges should better integrate socio-emotional skill development into their curricula and reward systems. Offering more experiential learning opportunities, such as group projects and leadership roles, can provide avenues for students to practice and demonstrate these skills. Additionally, adopting holistic evaluation systems that recognize communication, problem-solving, and collaboration would incentivize high school students and young adult learners to invest in their socio-emotional skill development.

Table 3: Descriptive Statistics, Socio-Emotional Skill Training Programs

	Overall	Disagreeableness	Conscientiousness	Emotional Stability	Extraversion	Openness	Multiple
Overall (N)	61	3	11	19	8	17	3
Grade level (%)							
Primary or less	35.71	50.00	33.33	15.79	42.86	58.33	100.00
Secondary	10.71	50.00	0.00	5.26	14.29	16.67	0.00
Post-secondary	53.57	0.00	66.67	78.95	42.86	25.00	0.00
Setting (%)							
Classroom	57.14	NA	50.00	NA	50.00	66.67	NA
Outside classroom	42.86	NA	50.00	NA	50.00	33.33	NA
Instructor (%)							
Teaching staff	43.75	50.00	50.00	NA	NA	NA	40.00
Other	56.25	50.00	50.00	NA	NA	NA	60.00
Targeting (%)							
General	50.00	NA	50.00	NA	50.00	50.00	50.00
Targeted	50.00	NA	50.00	NA	50.00	50.00	50.00
Technology (%)							
Technology-enhanced	14.58	50.00	0.00	10.00	16.67	6.67	33.33
Traditional	85.42	50.00	100.00	90.00	83.33	93.33	66.67

Note: This table reports descriptive statistics for the overall training effects extracted from meta-analyses of socio-emotional skill interventions. Each observation corresponds to an overall pooled training effect reported in a meta-analysis. Grade level indicates the educational stage of the primary target population of the interventions. Setting identifies the environment in which interventions were delivered. Instructor reports the primary implementer of the interventions. Targeting indicates whether interventions were designed for general or specific targeted populations. Technology identifies whether technology-assisted components were used. Percentages represent the share of overall training effects within each characteristic category.

Figure 4: Training Overall Impact on Big Five Personality Traits, RE, All Sample

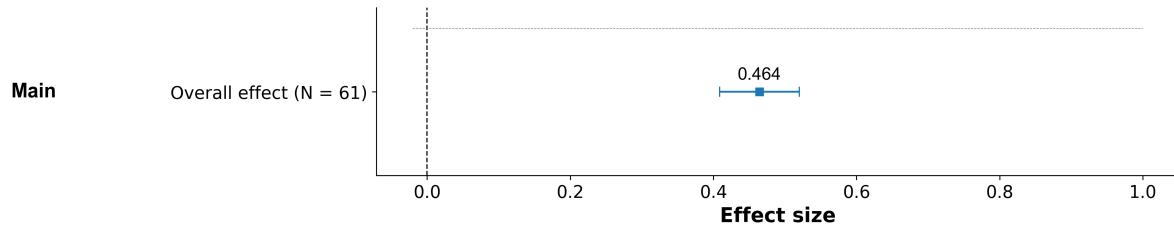
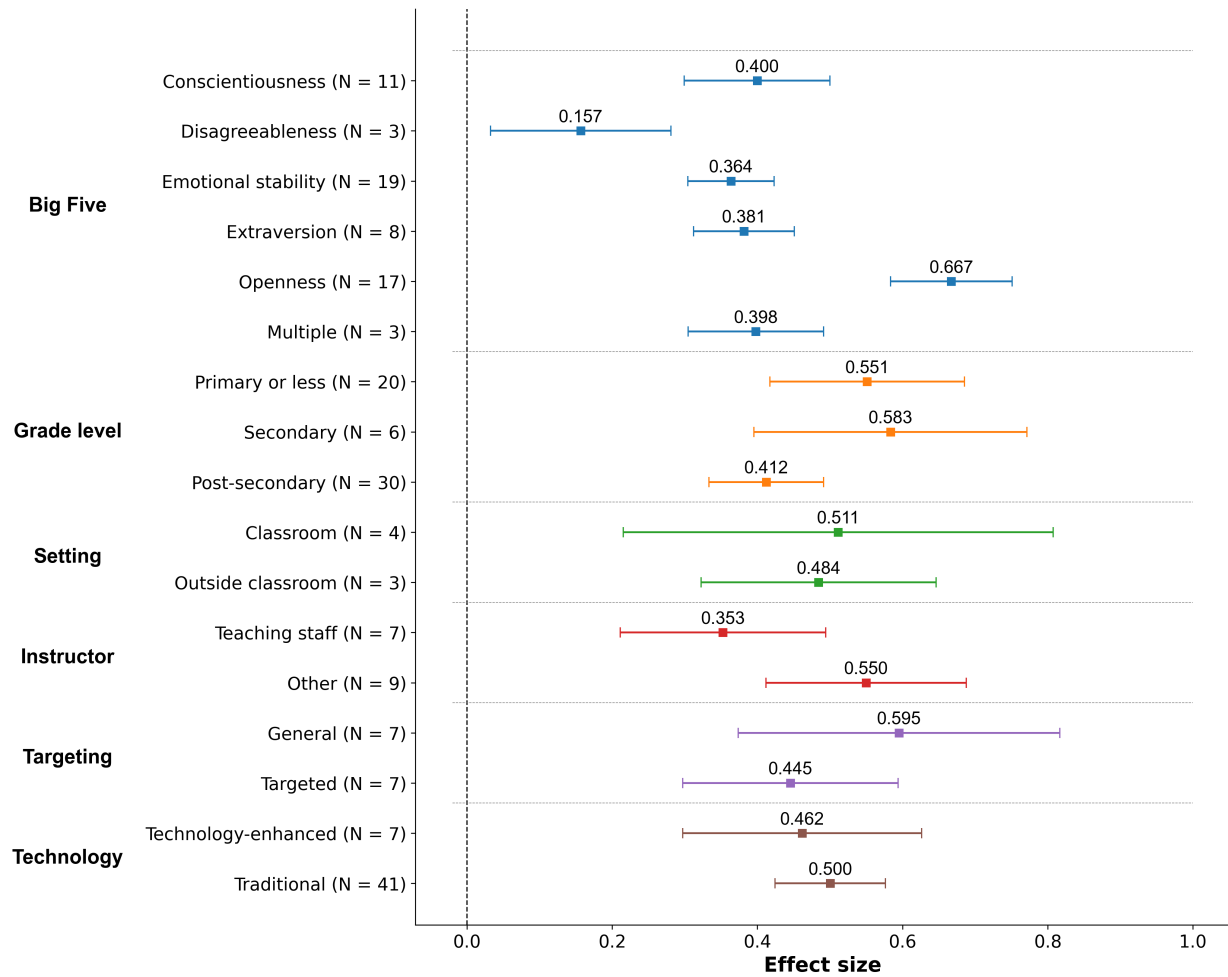


Figure 5: Heterogeneity of Training Impacts, RE, All Sample



Note: Figures 4 and 5 report pooled estimates of the effects of socio-emotional skill training programs on labor market outcomes. Each point estimate represents the average effect of participation in a training program, expressed in standard deviation units of the outcome. Estimates are based on a random-effects meta-analysis. Confidence intervals reflect 95 percent uncertainty bands. The total number of estimates included in each pooled effect is reported in parentheses.

Effects by Program Setting and Instructor

The setting in which socio-emotional skills are taught appears to have a minimal influence on training effectiveness, with both classroom-based programs ($\theta = 0.51$) and those conducted outside the classroom ($\theta = 0.48$) yielding moderate, yet insignificant effects. This suggests that the physical environment—whether in a traditional classroom or an alternative setting—does not drastically affect outcomes. Likewise, technology-enhanced programs ($\theta = 0.46$) perform similarly to traditional in-person approaches ($\theta = 0.50$), indicating that socio-emotional skills can be developed effectively regardless of whether the training is delivered face-to-face or through digital platforms.⁵

However, the instructor played a significant role in training effectiveness. Surprisingly, regular teaching staff ($\theta = 0.35$), such as teachers, were less effective at training socio-emotional skills compared to others, such as nonprofit trainers, psychologists, and life coaches ($\theta = 0.55$). These findings suggest that teaching staff often lack the consideration, expertise, or strategies needed to effectively develop students' or trainees' soft skills. It appears that regular educators need to place greater emphasis on socio-emotional skills and develop more effective approaches for teaching them. Inviting non-regular instructors into classrooms can be an option for more effective socio-emotional trainings.

Effects by Targeting Approach

General interventions ($\theta = 0.50$) tend to be more impactful than targeted programs ($\theta = 0.46$), but both approaches have unique roles in designing effective socio-emotional skills training interventions. While broader applicability often leads to higher, more consistent outcomes across diverse populations in general programs, making them ideal for large-scale interventions, targeted programs are indispensable for meeting the specific needs of certain high-risk groups. Universal programs may be more straightforward because they do not focus solely on individuals who may present more complex challenges and require addressing a range of external constraints such as socioeconomic challenges, disabilities, and family dynamics, which can limit the effectiveness of training (Piquero, et al, 2016).

However, it is crucial to emphasize that this finding does not discourage the use of targeted interventions; rather, it highlights the need for thoughtful consideration in their delivery to enhance effectiveness. Targeted programs, while sometimes facing challenges due to these external factors, are essential for addressing specific traits or the needs of particular groups. They allow for deeper, more focused improvements in socio-emotional skills, making them indispensable for specialized populations. Therefore, designing interventions that consider how to mitigate the dilution of focus and maximize impact, particularly in environments with numerous external challenges, is key. Combining general and targeted approaches strategically ensures that interventions are both broadly impactful and finely tuned to meet specialized needs.

Other Discussions

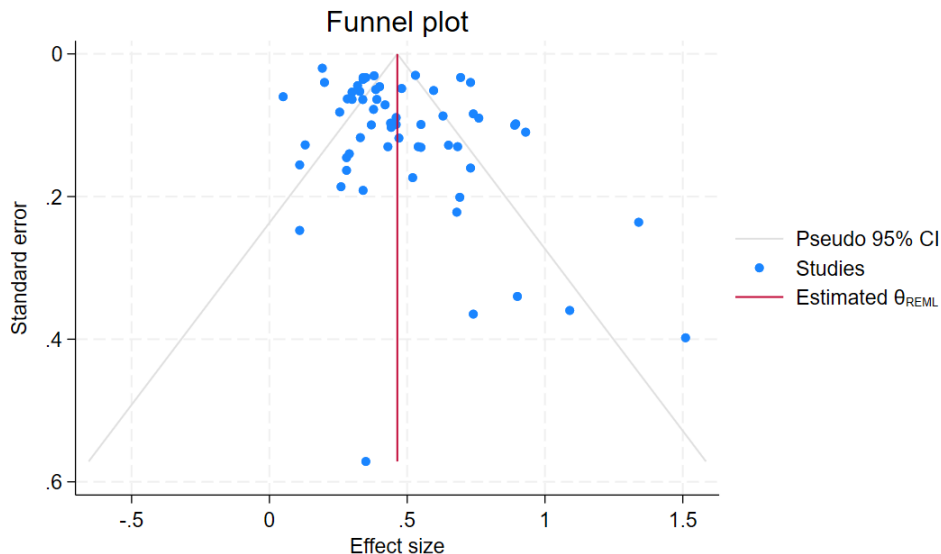
While it is uncommon in the meta-analytic literature to explore the impact of training pedagogy, there is some evidence that it plays a significant role in determining the effectiveness of

⁵The estimates for the program setting and instructor are with extremely small sample, causing large standard errors. The next version of paper will try to accommodate the sample size issue and reevaluate these estimates.

socio-emotional skill interventions. The heterogeneity observed across different pedagogical methods suggests that the choice of instructional approach can substantially influence outcomes. For example, Zhang et al. (2023) found that direct instruction was one of the most effective approaches for socio-emotional learning in virtual reality programs, even though it was one of the least utilized methods. This highlights the potential of direct, structured teaching to produce significant skill development when applied correctly. Additionally, inquiry-discovery methods also emerged as highly effective, fostering creativity and problem-solving by encouraging exploration and active engagement. Wang et al. (2024) similarly emphasize that inquiry-based learning is particularly useful in promoting divergent thinking and creativity. Meanwhile, mentor-based and collaborative learning methods demonstrated moderate effectiveness, highlighting the importance of personalized guidance and teamwork in skill development. In contrast, methods such as game-based learning exhibited lower impacts, though they may still enhance engagement and motivation.⁶

The funnel plot in Figure 6 shows a small hint of publication bias, potentially due to publication bias from each meta-analysis being carried over.

Figure 6: Training Estimation: Publication Bias



Note: This figure presents a funnel plot assessing potential publication bias in the estimated effects of socio-emotional skill training programs on labor market outcomes. Each point represents a study-level effect estimate plotted against its standard error.

7 Conclusion

In this study, we conduct a quantitative meta-analysis of recent empirical evidence on wage returns to socio-emotional skills, measured using the Big Five Inventory (BFI) personality traits and related keywords, as well as the impacts of training programs on socio-emotional skills development.

⁶It is important to note the higher effectiveness of specialized staff such as psychologists or life coaches as opposed to regular teachers in delivering non-cognitive skills training. This points to the importance to either offer additional training to teachers to deliver the relevant socio-emotional skills modules or to incorporate such specialized staff

As socio-emotional skills have become increasingly important for labor market success, the body of empirical research on these skills has expanded substantially over the past one to two decades. Against the backdrop of rapid technological shifts and growing labor market interest, this paper provides highly synthesized evidence on the labor market returns to, and training malleability of, socio-emotional skills.

We construct a highly synthesized database of relevant studies using a human-machine collaborative approach that combines large-scale web scraping with generative AI language model (GPT) information extraction and final human screening. Using this database, we implement a DerSimonian and Laird panel random effects (RE) estimation. One other methodological contribution of this paper lies in this efficient and replicable database construction protocols, which complements the paper’s substantive findings by offering a scalable framework for evidence synthesis in rapidly growing evidence.

Our findings yield several key takeaways. In the returns analysis, we find that a one standard deviation (SD) increase in each Big Five Inventory (BFI) trait is associated with a 1.4 to 3.1 percentage point increase in log real monthly wages. Although these magnitudes are less than one-third of the commonly reported returns to cognitive skills (approximately 10 percentage points), our robust estimates indicate that socio-emotional skills do generate meaningful wage returns in labor markets. Among the BFI traits, Disagreeableness yields the highest return (3.1 percentage points), followed by Emotional Stability and Openness. These results suggest that disagreeableness may play an important role in wage determination in contemporary labor markets that value leadership, competitiveness, and negotiation, while Openness and Emotional Stability reflect skills related to creativity, curiosity, and stress management.

We further find that training interventions targeting socio-emotional skills produce an average increase of 0.46 SD across BFI traits. Training effects are largest for attitudinal traits, particularly Openness (0.67 SD) and Conscientiousness (0.40 SD), while Disagreeableness exhibits the smallest training impact. Despite its high wage return, disagreeableness may be less socially preferred in certain cultural contexts, which could partially explain the limited training effects. Alternatively, the development of disagreeableness and other relational or emotional traits may require longer-term or more intensive interventions.

While cognitive skills are generally less malleable in adulthood, our findings suggest that socio-emotional skills remain comparatively malleable in adult populations, with moderately large training impacts observed in post-secondary settings. Moreover, non-regular teaching staff appear to be more effective in delivering socio-emotional skills training than regular teachers, while classroom settings and delivery modalities show limited differential effects.

Despite the highly synthesized measure and findings as well as methodological contributions, this study has several limitations. First, our narrow focus on log real monthly wages as the primary outcome may exclude other high-quality evidence capturing broader labor market effects. Second, the synthesis of training impacts relies exclusively on existing meta-analyses, which may warrant additional distributional analyses using alternative weighting schemes. Third, the relatively small sample size in the training estimations limits statistical power and calls for further sample expansion. We plan to address these limitations in future versions of the paper.

In addition, extending the analysis to alternative outcomes—such as employment, job stability, and entrepreneurship—may yield important insights into the broader economic benefits of socio-emotional skills beyond wage returns alone. From a methodological perspective, altering web-scraping and GPT tools may further optimize data extraction protocols.

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A Appendix

A.1 Web-scraping in Google Scholar

We collected research and policy publications released between 2019 and 2024 by web-scraping Google Scholar. Web-scraping was conducted using the Python package "Scholarly", with keyword searches implemented through Boolean operators. Through several pilot tests, six batches of keyword searches produced a large pool of literature on skill returns. For the analysis of training impacts, we mapped commonly used terms associated with training for each of the Big Five Inventory (BFI) traits—such as "creativity" training for openness, "communication" for extraversion, or "stress management" for emotional stability—to guide keyword selection.

We also tested several additional keywords, such as "leadership", "cooperation", "organized", "altruism", and "latent skills", to identify suitable supplementary measures of non-cognitive skills. When certain keywords retrieved unrelated studies (for example, the term "cooperation" yielded many papers on the benefits of economic cooperation in the context of bilateral trade agreements), these studies were excluded from the final list.

A.2 Information Extraction by the Generative Pre-Training Transformer (GPT)

To systematize the literature, after web-scraping the studies, we retrieved information from each study, including the title, authors, publication year, journal name, journal impact factor, number of citations, abstract, keywords, JEL codes, description of the data type, main methodology, whether the study conducted a causal analysis, dependent variables, non-cognitive skill indicators, other key independent variables, whether the study included training programs, main findings, and a link to the paper.

To enhance the efficiency of data collection, we trained OpenAI's GPT-4.0 to extract and summarize key information. GPT-4.0 generated very few errors when collecting structured information but occasionally produced inaccuracies when summarizing complex, qualitative, or lengthy content.⁷ All machine-generated errors were subsequently corrected through human review. This machine-assisted approach substantially improved the efficiency of our information collection process.

A.3 The Database Refinement

A.3.1 List of Final Papers on the Non-Cognitive Skill Returns

Figure A1 describes our searching, filtering, and coding approaches for the skill returns estimation. Boolean keyword searches were employed for web-scraping, combining terms related to wage returns, non-cognitive skills, and labor market outcomes. Google Scholar offered a credible volume of papers, so we scraped the database for the analysis and excluded all papers retrieved unrelated to the research.⁸

To ensure comparability and consistency, we refined the dataset through rigorous human screening, applying clear inclusion criteria. Papers were restricted to those written in English, reporting

⁷AI experts have suggested that OpenAI's GPT-based search may exhibit error rates of up to five percent. Recently developed tools, such as ImpactAI from the World Bank's Development Impact (DIME) group, and their broader applications and underlying algorithms, may further enhance this process once released.

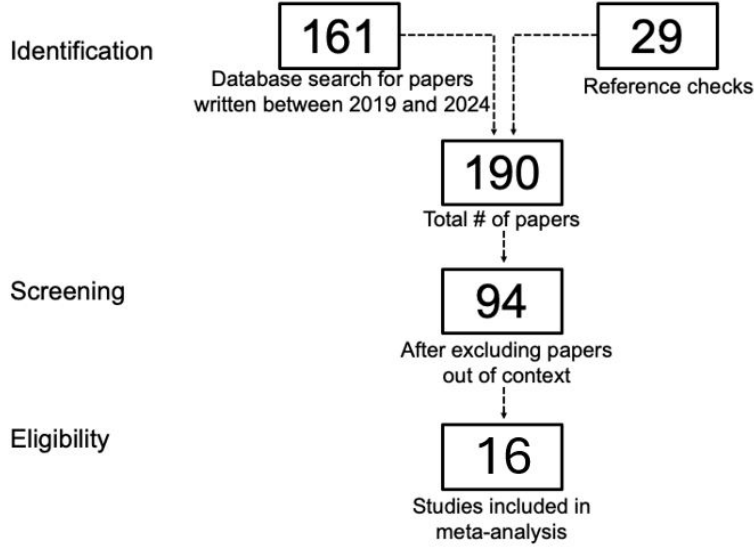
⁸Whereas the training analysis used systematic review papers only because individual papers report impacts with large heterogeneity. With this approach, we searched more repositories to collate as many systematic review papers.

Figure A1: The Literature Selection and Analysis Process

Search strategy How did we extract the papers?	Inclusion rules Which papers should be included?	Coding rules How did we classify the papers?
- Extracted papers from Google Scholar for the period 2019-2024 - Included papers found from looking up the references - Keyword searches using Boolean operator: = ("returns" OR "income" OR "wage" OR "earnings") AND ([vector of non-cognitive skill keywords]) AND ("labour market" OR "labor market" OR "employment")	- Papers written in English - Uses a continuous, standardized measure of non-cognitive skills. Papers using dummy variables to represent non-cognitive skills (e.g., training dummy) are excluded - Wage as the dependent variable - Estimation equation is expressed as semi-log equation, such that the estimated effect is in terms of % increase in wage for a 1 SD change in the non-cognitive skill - Studies from all geographic regions, industries and occupations - No restriction on the specific Big Five inventory	- Collected estimates for the main model along with heterogeneity and robustness checks - Identified the quality of evidence, particularly whether an IV is used - Identified the potential moderators: - Control for educational attainment - Control for cognitive skills - Gender - Region/continent

wage as the dependent variable, and using a continuous, standardized measure of non-cognitive skills (e.g., a 1 SD increase). Studies using dummy variables for non-cognitive skills were excluded. Additionally, the estimation equations needed to follow a semi-log form, reporting percentage changes in wages. To enhance comparability further, we restricted the analysis to studies reporting impacts in terms of SDs, a consistent metric that allows results to be compared on the same scale. This additional refinement reduced the dataset to 16 papers with 613 effect sizes, including 568 for non-cognitive skills and 45 for cognitive skills. Focusing on SD impacts simplifies interpretation and minimizes inconsistencies arising from varying measurement units or methods across studies. The PRISMA chart is shown in Figure A2, and the final list of papers is in Table A1.

Figure A2: PRISMA Chart: Returns to Non-Cognitive Skills



Finally, we applied coding rules to extract consistent and detailed data, such as estimates from the main models, heterogeneity checks, and robustness analyses. We classified study quality as follows: 1 = RCT/Experimental design, 2 = Quasi-experimental (DiD, RDD, IV, Matching, etc.), 3 = Correlational evidence, 4 = Qualitative evidence. Figure A3 shows the hierarchy of rigor we employed. Also, we identified potential moderators like educational attainment, cognitive skill controls, gender, and geographic region.

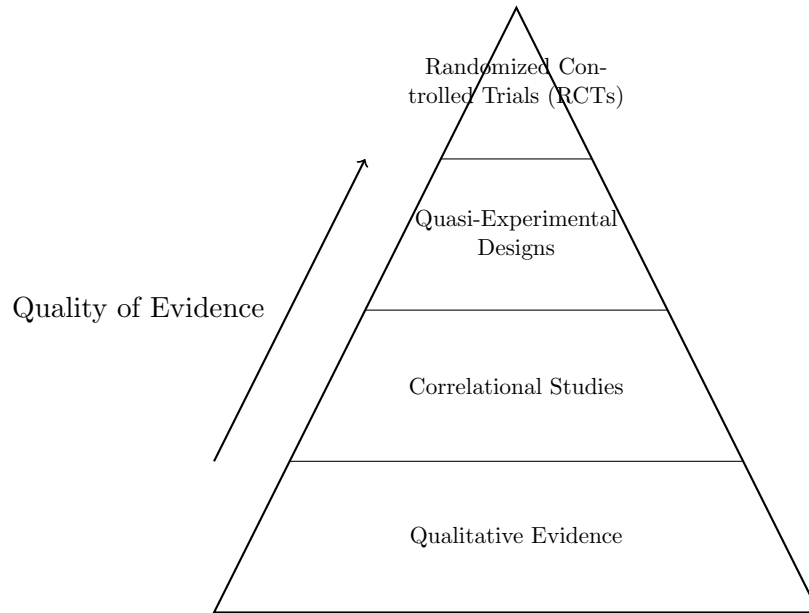
Leveraging the large number of observations, we were able to subsample the evidence by education controls, gender, and country income group. Countries were classified using the 2024 World Bank Group Country Classification by Income, which categorizes economies into low-income, lower-middle-income, upper-middle-income, and high-income groups.⁹ However, due to limited sample sizes within income groups, we eventually collapsed the classification into two groups: high-income countries, representing developed economies, and developing countries, comprising all remaining income categories.

A.3.2 List of Final Papers on the Program Intervention

Figure A4 summarizes our approach to searching, filtering, and coding studies on program interventions. Given the vast and rapidly expanding body of primary evidence on training effectiveness, we focused our search strategy on identifying and synthesizing existing meta-analyses. This approach is not only practical—given the substantial time and resources required to systematically screen and analyze thousands of individual studies—but also methodologically advantageous. By conducting a meta-meta-analysis, we synthesize evidence across diverse contexts and study designs at a higher level of aggregation. This allows us to draw on comprehensive findings, identify broader

⁹Income group thresholds are defined by GNI per capita of \$1,135 or less for low-income countries, \$1,136–\$4,465 for lower-middle-income countries, \$4,466–\$13,845 for upper-middle-income countries, and \$13,846 or more for high-income countries.

Figure A3: The Evidence-Based Pyramid



patterns that are robust across bodies of synthesized evidence, and assess variation in program characteristics such as duration, delivery setting, and target populations.

To identify relevant studies, we implemented a Boolean keyword search in Google Scholar. Because training programs often do not explicitly reference the Big Five personality traits, we mapped each trait to commonly used terms in the training literature. Figure [A4](#) displays the specific keywords employed in this mapping process.

Figure A4: Keywords Mapped to Each Big Five Personality Trait

Search strategy How did we extract the papers?	Inclusion rules Which papers should be included?	Coding rules How did we classify the papers?
- Mapping each Big Five Personality Trait to commonly used terms in training interventions (e.g., “creativity training” for openness, “stress management” for emotional stability) - Extracted papers on meta-analysis of training effectiveness from Google Scholar - Search yielded 3,194 results - Keyword searches using Boolean operator: = “meta-analysis” AND (“creativity” AND (“training” OR “intervention” OR “program”) AND (“school” OR “students” OR “adolescents”))	- Papers written in English - Only meta-analyses are included, - Focus on school- or work-based interventions; excluded if family- or home-based - Excludes specific populations (e.g., individuals with disabilities, athletes). - Narrowed down to a final list of 29 meta-analyses	- Collected estimates for the main model along with heterogeneity estimates - Identified the potential moderators: - Grade level - Intervention setting - Instructor - Targeting - Technology use - Duration - Time frame for evaluation

The initial search yielded 3,194 results, many of which were duplicates or irrelevant to the research objectives. We then applied strict inclusion criteria, restricting the sample to English-language meta-analyses that focused on school- or work-based training interventions. Studies examining home-based approaches or targeting specific populations, such as individuals with disabilities or athletes, were excluded. This rigorous screening process narrowed the dataset to a final set of 29 meta-analyses aligned with our focus on interventions designed to enhance non-cognitive skills. From these meta-analyses, we extracted 364 estimates from the main specifications, along with heterogeneity estimates to capture variation in training effectiveness. The PRISMA flow diagram is shown in Figure A5, and the list of included papers is reported in Table A2.

Figure A5: PRISMA chart: Training Programs



We further applied a structured coding framework to identify potential moderators that could influence the success of training programs. These included factors such as the grade level or age of participants, the setting of the intervention (e.g., school or workplace), the qualifications of the instructors, whether the training was universally applied or targeted specific groups, the use of technology, the duration of the intervention, and the time frame for evaluating outcomes. By exclusively focusing on meta-analyses, combining targeted search strategies, and employing detailed coding rules, our methodology ensures a robust and comprehensive assessment of how training programs impact the development of non-cognitive skills associated with the BFI Traits.

A.4 Additional Tables and Figures

Table A1: Filtered List of Literature in Our Non-Cognitive Skills Database

No.	Author	Location	Income Group	Method	Type	Measure
1	Acosta et al. (2015)	Colombia	Developing	2	Big Five/Others	SD
2	Cunningham and Villaseñor (2016)	Peru	Developing	3	Big Five/Others	SD
3	Deming (2017)	USA	High income	3	Big Five/Others	SD
4	Díaz et al. (2012)	Peru	Developing	2,3	Big Five/Others	SD
5	Fletcher (2013)	USA	High income	2,3	Big Five/Others	Continuous
6	Glewwe et al. (2022)	China	Developing	3	Big Five/Others	SD
7	Heineck and Anger (2010)	Germany	High income	2,3	Big Five/Others	Dummy
8	Heineck (2011)	UK	High income	2,3	Big Five/Others	Dummy
9	Hilger et al. (2022)	Bangladesh	Developing	2,3	Big Five/Others	Continuous
10	Laible and Brenzel (2022)	Germany	High income	3	Big Five/Others	SD
11	Lavado et al. (2022)	Peru	Developing	3	Big Five/Others	Continuous
12	Lindqvist and Vestman (2011)	Sweden	High income	2	Big Five/Others	SD
13	McLean et al. (2019)	Canada	High income	3	Big Five/Others	SD
14	Otchia and Yamada (2019)	Ghana	Developing	2	Big Five/Others	SD
15	Palczyńska (2021)	Poland	High income	3	Big Five/Others	SD
16	Rozhkova et al. (2021)	Russia	Developing	3	Big Five/Others	SD

Table A2: Overview of Meta-Analyses on Non-Cognitive Skills Training

No.	Author	Type	Specific skill	School level*
1	Torgal et al. (2021)	Agreeableness	Empathy	ES, HS
2	Polanin et al. (2012)	Agreeableness	Empathy	ES, HS
3	Dignath et al. (2008)	Conscientiousness	Self-regulation	ES
4	Epton et al. (2017)	Conscientiousness	Goal-setting	Not mentioned
5	Frieese et al. (2017)	Conscientiousness	Self-control	Out of school
6	Li et al. (2021)	Conscientiousness	Self-control	PE, ES, HS
7	Piquero et al. (2016)	Conscientiousness	Self-control	PE, ES
8	Theobald (2021)	Conscientiousness	Self-regulation	PS
9	Dawson et al. (2020)	Emotional Stability	Mindfulness training	PS
10	Howard and Gutworth (2020)	Emotional Stability	Social skills	NA
11	Kraag et al. (2006)	Emotional Stability	Coping skills	ES
12	Moltrecht et al. (2021)	Emotional Stability	Emotional regulation	ES, HS
13	Kermani et al. (2022)	Extraversion	Communication skills	PS
14	Luo et al. (2022)	Extraversion	Socio-emotional competence	PE
15	McEwan et al. (2017)	Extraversion	Teamwork	NA
16	Murano et al. (2020)	Extraversion	Social and communication skills	PE
17	Shin and Lee (2021)	Extraversion	Pro-social behavior	ES, HS
18	Zhang et al. (2024)	Extraversion	Social interaction skills	PE, ES, HS
19	de Mooij et al. (2020)	Multiple	Inter-personal and emotional skills	ES, HS
20	Ang et al. (2022)	Openness	Resilience	PS
21	Bi et al. (2020)	Openness	Scientific creativity	PE, ES, HS, PS
22	Calo et al. (2024)	Openness	Resilience	PS
23	Ejaz et al. (2024)	Openness	Resilience	PS
24	Huang (2005)	Openness	Creativity	ES, HS
25	Kwon et al. (2016)	Openness	Creativity	ES, HS
26	Leppin et al. (2014)	Openness	Resilience	PS
27	Ma (2006)	Openness	Different skills	HS, PS
28	Sio and Lortie-Forgues (2024)	Openness	Creativity	ES, PS
29	Wang et al. (2024)	Openness	Creativity	PE, ES, HS, PS

* Note: Abbreviations: PE = pre-elementary, ES = elementary; HS = high school; PS = post-secondary.

Table A3: Descriptive Statistics, Labor Returns Analysis by Region

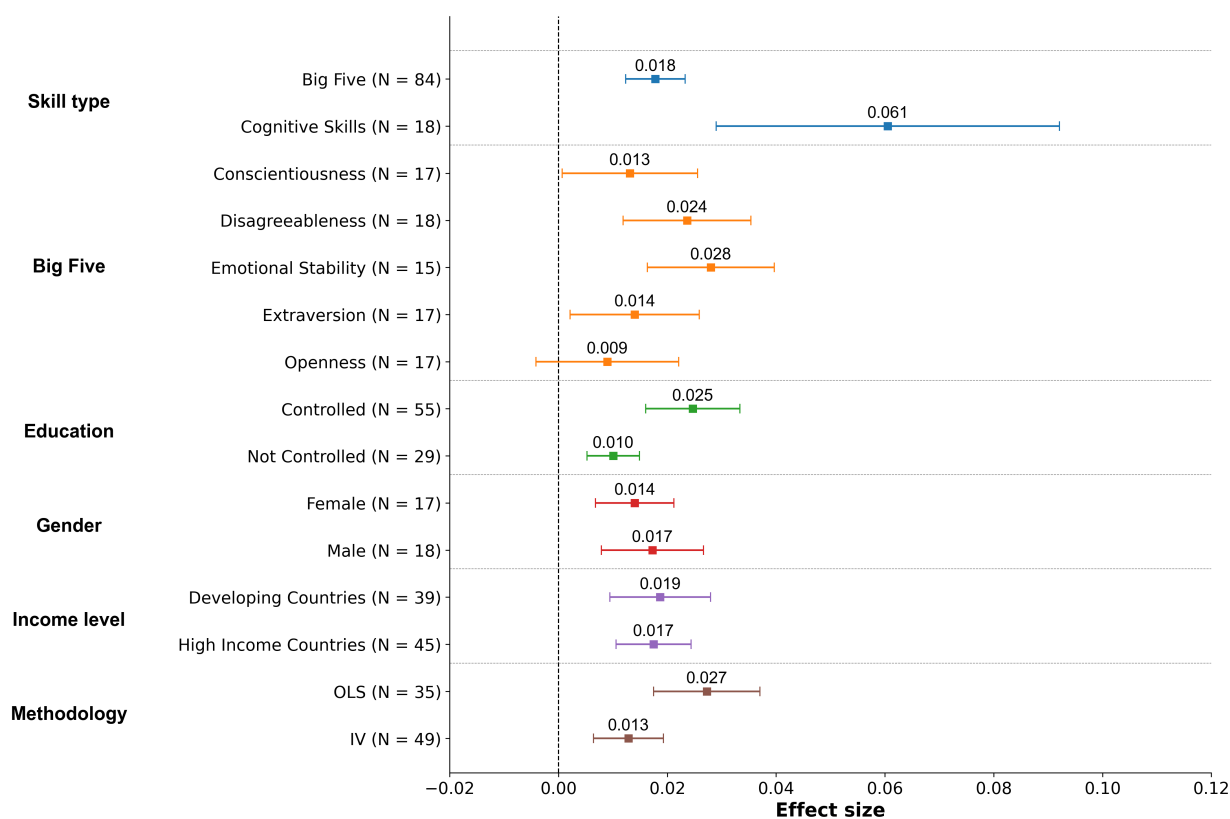
	Overall	Africa	Asia	Europe	Latin America	North America
Number of estimates	613	19	88	337	107	62
Estimate type (%)						
Big Five	92.7	78.9	96.6	95.3	90.7	80.6
Cognitive Skills	7.3	21.1	3.4	4.7	9.3	19.4
Education control (%)						
Yes	47.0	52.6	98.9	33.8	45.8	45.2
No	53.0	47.4	1.1	66.2	54.2	54.8
Gender (%)						
Female	22.7	0.0	0.0	39.8	4.7	0.0
Male	22.8	0.0	0.0	40.1	4.7	0.0
Mixed	54.5	100.0	100.0	20.2	90.7	100.0
Methodology (%)						
OLS	28.5	0.0	2.3	40.9	23.4	16.1
IV	71.5	100.0	97.7	59.1	76.6	83.9

Note: This table reports descriptive statistics for the 613 wage-return estimates used in the meta-analysis of labor market returns to socio-emotional skills. The unit of observation is the individual coefficient. The regions are based on the World Bank region classification at the time of the study. Percentages represent the share of estimates within each characteristic category (estimate type, education control, gender, and methodology).

Figure A6: Big Five Personality Traits Questionnaire

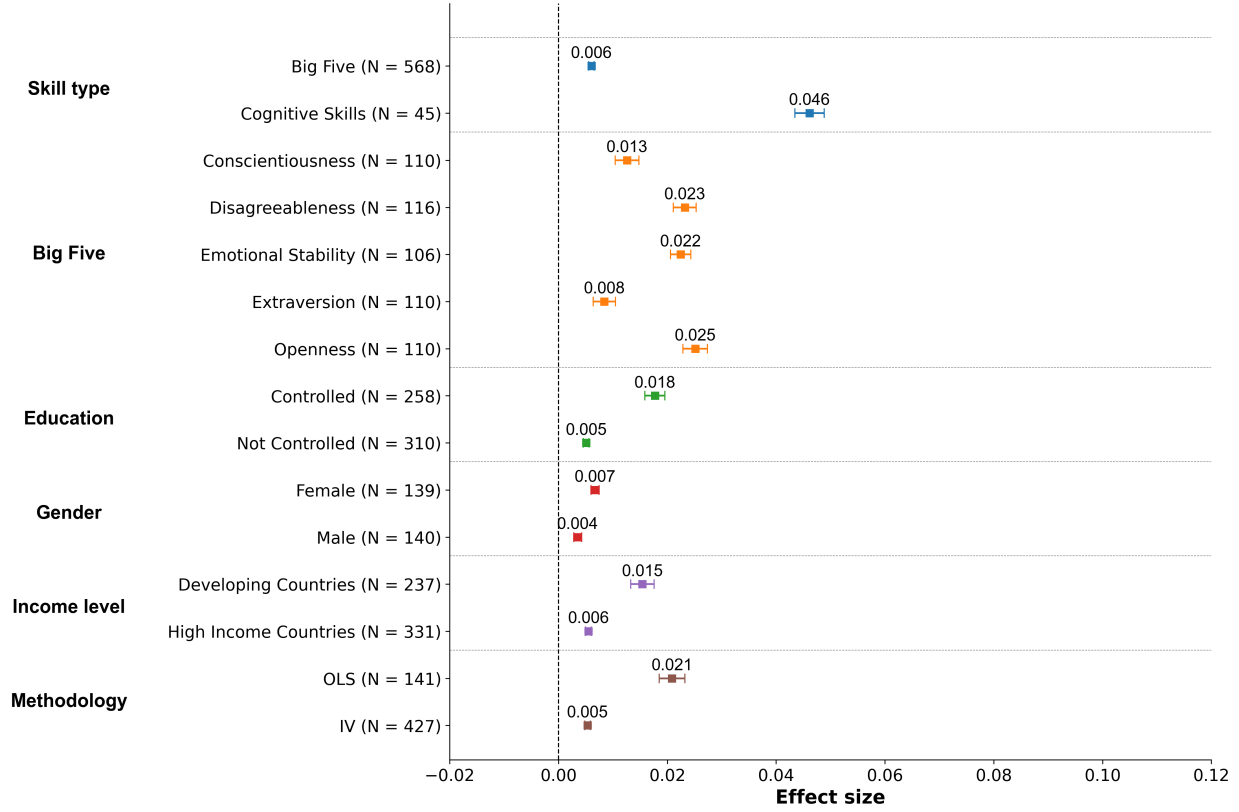
The Big Five Inventory (BFI)				
Here are a number of characteristics that may or may not apply to you. For example, do you agree that you are someone who likes to spend time with others? Please write a number next to each statement to indicate the extent to which you agree or disagree with that statement.				
Disagree strongly 1	Disagree a little 2	Neither agree nor disagree 3	Agree a little 4	Agree Strongly 5
<u>I see Myself as Someone Who...</u>				
___ 1. Is talkative				___ 23. Tends to be lazy
___ 2. Tends to find fault with others				___ 24. Is emotionally stable, not easily upset
___ 3. Does a thorough job				___ 25. Is inventive
___ 4. Is depressed, blue				___ 26. Has an assertive personality
___ 5. Is original, comes up with new ideas				___ 27. Can be cold and aloof
___ 6. Is reserved				___ 28. Perseveres until the task is finished
___ 7. Is helpful and unselfish with others				___ 29. Can be moody
___ 8. Can be somewhat careless				___ 30. Values artistic, aesthetic experiences
___ 9. Is relaxed, handles stress well				___ 31. Is sometimes shy, inhibited
___ 10. Is curious about many different things				___ 32. Is considerate and kind to almost everyone
___ 11. Is full of energy				___ 33. Does things efficiently
___ 12. Starts quarrels with others				___ 34. Remains calm in tense situations
___ 13. Is a reliable worker				___ 35. Prefers work that is routine
___ 14. Can be tense				___ 36. Is outgoing, sociable
___ 15. Is ingenious, a deep thinker				___ 37. Is sometimes rude to others
___ 16. Generates a lot of enthusiasm				___ 38. Makes plans and follows through with them
___ 17. Has a forgiving nature				___ 39. Gets nervous easily
___ 18. Tends to be disorganized				___ 40. Likes to reflect, play with ideas
___ 19. Worries a lot				___ 41. Has few artistic interests
___ 20. Has an active imagination				___ 42. Likes to cooperate with others
___ 21. Tends to be quiet				___ 43. Is easily distracted
___ 22. Is generally trusting				___ 44. Is sophisticated in art, music, or literature
Scoring:				
BFI scale scoring ("R" denotes reverse-scored items):				
Extraversion: 1, 6R, 11, 16, 21R, 26, 31R, 36				
Agreeableness: 2R, 7, 12R, 17, 22, 27R, 32, 37R, 42				
Conscientiousness: 3, 8R, 13, 18R, 23R, 28, 33, 38, 43R				
Neuroticism: 4, 9R, 14, 19, 24R, 29, 34R, 39				
Openness: 5, 10, 15, 20, 25, 30, 35R, 40, 41R, 44				

Figure A7: The Wage Returns to Big Five Personality Traits, RE, Literature Main Results



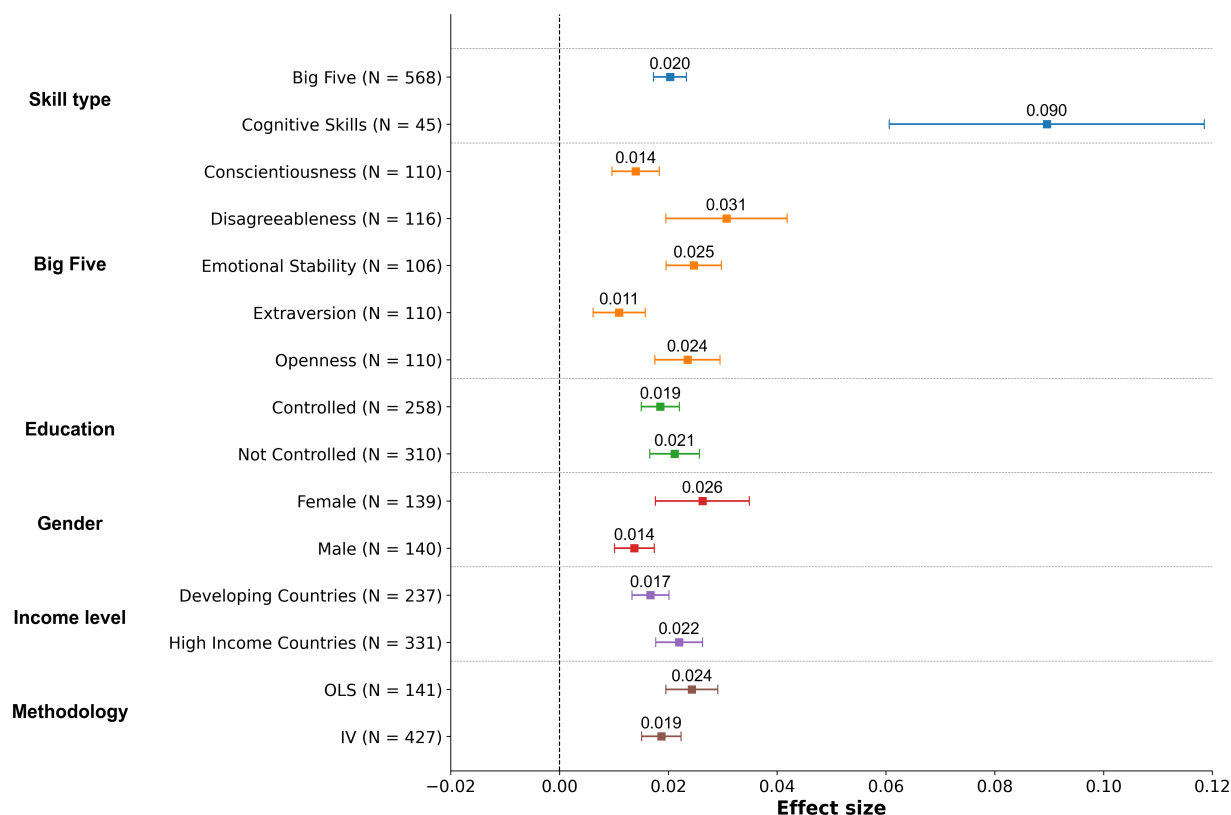
Note: This figure reports pooled estimates of the labor market returns to socio-emotional skills, measured using the Big Five personality traits, restricted to the main specifications reported in the studies. Each point estimate represents the average percentage change in real wages associated with a one standard deviation (SD) increase in the corresponding skill. The estimates are based on a DerSimonian–Laird random-effects meta-analysis. Confidence intervals reflect 95 percent uncertainty bands. The total number of estimates included in each pooled effect is reported in parentheses. All estimates are drawn from studies that use log real wages as the dependent variable.

Figure A8: The Wage Returns to Big Five Personality Traits, FE, All Sample



Note: This figure reports pooled estimates of the labor market returns to socio-emotional skills, measured using the Big Five personality traits. Each point estimate represents the average percentage change in real wages associated with a one standard deviation (SD) increase in the corresponding skill. The estimates are based on a fixed-effects meta-analysis. Confidence intervals reflect 95 percent uncertainty bands. The total number of estimates included in each pooled effect is reported in parentheses. All estimates are drawn from studies that use log real wages as the dependent variable. The total number of Big Five estimates is 568, whereas the sum of estimates across individual Big Five traits is 552. The difference (16 estimates) reflects results from one study that reports interaction effects between Big Five traits and tenure, which are included in the overall count.

Figure A9: The Wage Returns to Big Five Personality Traits, RE, Studies Controlling for All Big Five Personality Traits



Note: This figure reports pooled estimates of the labor market returns to socio-emotional skills, measured using the Big Five Personality Traits, restricted to studies that control for all Big Five traits. Each point estimate represents the average percentage change in real wages associated with a one standard deviation (SD) increase in the corresponding skill. The estimates are based on a DerSimonian–Laird random-effects meta-analysis. Confidence intervals reflect 95 percent uncertainty bands. The total number of estimates included in each pooled effect is reported in parentheses. All estimates are drawn from studies that use log real wages as the dependent variable. The total number of Big Five estimates is 568, whereas the sum of estimates across individual Big Five traits is 552. The difference (16 estimates) reflects results from one study that reports interaction effects between Big Five traits and tenure, which are included in the overall count.